

Project Challenge

Overview

This project challenge is designed to evaluate your ability to:

- Translate a business problem into an architecture and implementation plan.
- Design and build an agentic solution on the phData Intelligence Platform (see below) and a modern cloud/LLM stack.
- Communicate how the technical system works, clearly, accurately, and convincingly, to both technical and non-technical people.

You should treat this as a real client engagement where you are responsible for the technical approach: designing the system, building a credible proof-of-concept, and explaining how and why it works. The emphasis is on engineering depth and your ability to walk others through the system you built: the architecture, the trade-offs, the agent behavior, and how it stays safe and reliable.

The phData Intelligence Platform

The phData Intelligence Platform is an AI-native operating layer that sits on top of a modern data stack (e.g., Snowflake, cloud warehouses, event streams, SaaS systems). It:

- Connects enterprise data, knowledge, and processes through semantic models and shared services.
- Exposes a common foundation for agents (retrieval, tools, orchestration, governance, observability).
- Integrates with operational systems (ITSM, CRM, ERP, collaboration tools) so agents can both reason over context and take real actions (create/update tickets, trigger workflows, send notifications, etc.).

For this challenge, assume:

- The client's core data and content (tickets, assets, knowledge base, logs) are accessible through the Intelligence Platform as structured tables and indexed documents.
- The platform provides standard building blocks for:
 - Retrieval (RAG) over documentation and runbooks.
 - Access to structured IT data (tickets, assets, SLAs, telemetry).
 - Tooling endpoints to create/update incidents, post comments, and trigger simple automations.

You will design and build agents that live on this platform and orchestrate work across these capabilities. For the POC, you can simulate these platform building blocks locally with the data provided. You do not need access to the real platform.

Scenario

Client: ServeWell Hospitality

Industry: Multi-brand restaurant & retail (franchise + corporate locations)

Challenge: ServeWell operates 2,500+ locations across multiple brands. Their L1 IT support organization is under significant strain:

- **Slow L1 resolution times:** Store managers wait on hold or raise tickets for common issues ("POS won't start," "Wi-Fi down," "kitchen printer offline"). The average time to first meaningful response is 30–45 minutes during peak hours.
- **High volume of repetitive incidents:** 60–70% of tickets are known issues that map to existing runbooks (restart sequences, config checks, simple network resets) but still require human triage and step-by-step guidance.
- **Inconsistent quality across shifts and locations:** Some locations get quick, accurate help; others bounce between L1 and L2 due to incomplete information, missing logs, or poorly filled tickets.
- **Fragmented knowledge and tooling:**
 - Runbooks and SOPs live across Confluence, SharePoint, PDFs, and old email threads.

- Ticket history is in an ITSM tool (e.g., ServiceNow, Jira Service Management).
- Asset metadata (POS, routers, kiosks) is in a separate CMDB and inventory system.
- Agents and analysts constantly swivel-chair between these systems.

Leadership has three clear business pain points:

- Store downtime directly reduces revenue and customer satisfaction.
- L1 support cost and burnout are rising, with high attrition in the support center.
- L2 engineers are overloaded with basic tickets that should never reach them.

What ServeWell Wants from phData

ServeWell has engaged phData to design and prove out an IT Support Agent Suite on the phData Intelligence Platform that can:

- Triage and enrich incoming incidents using relevant content
- Improve routing and reporting
- Guide and/or resolve common L1 issues
- Propose next-best actions or trigger pre-approved automations
- Summarize and communicate clearly
- Operate safely and visibly, with guardrails and full auditability

ServeWell wants to start small but think big:

- Begin with one or two high-frequency incident types (e.g., “POS offline at store,” “receipt printer not working”).
- Prove the value in a narrow, realistic pilot that runs on the phData Intelligence Platform.
- Build toward a reusable accelerator that can be rolled out across more brands, regions, and IT domains.

Your Challenge

ServeWell has asked phData to propose and demonstrate how an IT Support Agent Suite on the phData Intelligence Platform could address this problem. Your task is to:

1. **Define the solution and architecture**

- a. Design an end-to-end agentic solution on the phData Intelligence Platform that addresses ServeWell's L1 IT support challenges. Your design should make clear:
 - i. The agent(s) and their responsibilities, and how work is orchestrated across them.
 - ii. How retrieval (RAG) over runbooks/knowledge base is structured, and how the agent grounds its responses.
 - iii. How the agent accesses structured IT data (tickets, assets, SLAs, telemetry) and when it calls tools vs. reasons.
 - iv. The actions the agent can take (e.g., create/update tickets, post comments, trigger automations) and the human-in-the-loop and guardrail points around them.
 - v. Your evaluation, guardrail, and observability strategy — how you'd know the agent is accurate, safe, and reliable before and after deployment.
 - vi. The key technical trade-offs you made and why (model choice, orchestration framework, retrieval approach, where to draw the autonomy line, etc.).

2. **Implement a proof-of-concept agent (or small set of agents) for a narrow slice of the problem**

- a. You are not expected to build a full production system, but you should implement a minimal, credible POC that demonstrates how an agent running "on" the Intelligence Platform would behave for one common incident type. For this challenge, you should consider developing the PoC with synthetic data on a local system. A strong POC:
 - i. Ingests or accepts a realistic incoming incident and triages/enriches it.

- ii. Retrieves and grounds against the relevant runbook(s) and structured data (asset, ticket history, SLA).
- iii. Walks toward a resolution — guiding steps and/or proposing/triggering a (simulated) action — and produces a clear summary back to the requester and the ticket.
- iv. Demonstrates at least one guardrail or evaluation check (e.g., refusal/escalation when confidence is low, an output check, or a hallucination/grounding check).
- v. Is organized as clean, readable, well-structured code with enough documentation (a README) that someone else could run it and understand the design.
- vi. Can intake fresh data and provide quality outputs.

3. Present how this solution delivers business ROI to a mixed business + technical audience

- a. Prepare a technical walkthrough and present it live. The audience is mixed: technical reviewers who will probe the implementation, and a non-technical stakeholder who needs to understand what the system does and trust it. Your walkthrough should:
 - i. Explain the architecture and how the pieces fit together, at the right level for each part of the audience.
 - ii. Demo the POC end-to-end on a real example, narrating what the agent is doing and why at each step.
 - iii. Show how you reason about correctness, safety, and reliability (evaluation, guardrails, observability) — and be candid about limitations and what you'd do next.
 - iv. Make the technical system understandable: a good walkthrough lets a non-technical stakeholder follow the value and a technical reviewer trust the engineering.
 - v. Intake new data that will run through the system successfully.

4. Test your system on unseen data

- a. Your panel of reviewers will provide you with a sample of test cases to feed into your system to test system design. Be prepared to walk through the design of your solution and be able to defend your choices, even if a test case does not run perfectly.

What we care about most: We are not grading you on a leaderboard score. We are evaluating how you **reason**, how you translate an ambiguous business problem into an architecture, the trade-offs you make and can defend, how you know your system is correct and safe, and how clearly you can explain all of this to both technical and non-technical people. A smaller, honest POC that you can defend deeply will beat a broad one you can only describe superficially. **Be prepared to explain every decision and to tell us, candidly, what you would improve with more time.**

Role Play: As you pitch, a small group of phData team members (3-4) will play the role of the client, and we'll ask some challenging, collaborative questions to see how you approach problem-solving. Here are the roles within the presentation:

- **Business Unit Vice President** - this role has the problem that needs to be solved, and the budget that will be used, but is not technical.
- **Director of Technology** - this role is responsible for ensuring the vendor that is selected delivers on solving the problem.

Reach out to your Recruiter for any questions or clarification along the way. Your recruiter will schedule time with the team when you're ready to present.

Best of luck!